

Minimal Point-Valued Stochastic Dynamics with Relaxing Memory

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We define a minimal point-valued stochastic dynamics on an abstract continuous state space Ω in which a visible trajectory x_n is coupled to an explicit exponentially relaxing memory density ρ_n . The projected process x_n is generally non-Markovian, whereas the augmented state $\sigma_n = (x_n, \rho_n)$ is Markov by construction. The memory update compresses ordered histories into a current density and thereby defines an intrinsic arrow of update order without assuming spacetime, physical time, or thermodynamic irreversibility. Coarse-graining on the memory-persistence scale gives an effective parameter $t = \alpha n$ and, when a separation of scales exists, an approximate SDE/Fokker–Planck description. Within this effective regime we define metastable dynamical knots and coordinate-robust relaxation diagnostics. Any calibration to physical units, relativistic dynamics, or particle-physics structure is left to later work.

I. INTRODUCTION

Many approaches to fundamental physics begin with structures such as spacetime, locality, fields, or quantum states [1–5]. Here we ask a more limited question: what is obtained if one assumes only a sequence of non-trivial update events, a visible state, and an internal memory that relaxes but is not externally prescribed?

The construction is close in spirit to random processes with reinforcement and self-interacting walks or diffusions biased by their visitation history, including the true self-avoiding walk, self-interacting diffusions, and self-repelling Brownian polymers [6–11]. Its distinctive feature is that the retained history is not a full occupation measure or local time, but an explicit exponentially relaxing density, in the spirit of reduced memory variables in nonequilibrium dynamics [12, 13]. This gives a finite Markov embedding while preserving non-Markovian visible dynamics.

The purpose of this paper is correspondingly narrow. We define the model, state the Markov embedding, extract the intrinsic coarse-grained time scale, and introduce metastability diagnostics for localized structures generated by the memory coupling. The paper does not assume a physical spacetime metric, conserved quantities, quantum postulates, or calibrated mass units.

II. MODEL AND MARKOV EMBEDDING

Let Ω be a continuous state space with a reference measure dx and local coordinates sufficient to write integrals and derivatives. The background coordinates are a mathematical scaffold for the generator and diagnostics, not a postulated physical spacetime. At update step n the full state is

$$\sigma_n = (x_n, \rho_n), \quad (1)$$

where $x_n \in \Omega$ is the visible state and ρ_n is a normalized non-negative memory density on Ω . The normalization is a baseline convention: it fixes the overall memory scale

and makes $M_n(K)$ below a visitation weight, while the same mechanism could be formulated with finite positive measures and a corresponding rescaling of η .

The visible update is

$$x_{n+1} = x_n + \varepsilon \xi_n - \eta \nabla \Phi_n(x_n), \quad (2)$$

where ξ_n is an independent isotropic unit-variance noise variable [14]. The memory evolves as

$$\rho_{n+1}(x) = (1 - \alpha)\rho_n(x) + \alpha G_\sigma(x - x_{n+1}), \quad 0 < \alpha \leq 1, \quad (3)$$

with G_σ a smooth normalized kernel of width σ . The memory-induced potential is

$$\Phi_n(x) = \int_{\Omega} K(x, y) \rho_n(y) dy. \quad (4)$$

The gradient in Eq. (2) acts on the first argument of K :

$$\nabla \Phi_n(x) = \int_{\Omega} \nabla_x K(x, y) \rho_n(y) dy, \quad (5)$$

whenever differentiating under the integral is justified. For a translation-invariant kernel $K(x, y) = K_\ell(x - y)$ this is the convolution of ρ_n with ∇K_ℓ .

Given σ_n and a fresh draw ξ_n , Eqs. (2)–(3) determine the distribution of σ_{n+1} . Thus the sequence $\{\sigma_n\}_{n \geq 0}$ is Markov. The visible sequence $\{x_n\}_{n \geq 0}$ alone is generally non-Markovian, because identical visible states can have different retained memory fields and hence different next-step laws. This is the basic Markov embedding of the model.

Equation (3) also explains the arrow of update order. It maps an ordered trajectory history to a current density by exponentially suppressing older contributions. For fixed x_{n+1} the affine update can be inverted as a map in ρ_n , but the inverse does not recover the discarded ordered sequence of past positions. The irreversibility used here is therefore history compression, not thermodynamic entropy production.

Iterating Eq. (3) gives, for $n > k$,

$$\rho_n(x) = (1 - \alpha)^{n-k} \rho_k(x) + \alpha \sum_{j=k+1}^n (1 - \alpha)^{n-j} G_\sigma(x - x_j). \quad (6)$$

Thus a contribution of age $m = n - j$ enters with weight $\alpha(1 - \alpha)^m \simeq \alpha e^{-\alpha m}$ for $\alpha \ll 1$. The persistence scale is of order α^{-1} update steps.

III. EFFECTIVE TIME

The update index n is not physical time. Nevertheless, the memory relaxation rate supplies an internal coarse-graining scale. When observables vary slowly on the persistence scale, it is natural to define the dimensionless parameter

$$t = \alpha n. \quad (7)$$

This parameter counts persistence-scale units along the update sequence. In a joint limit $\alpha \rightarrow 0$ and $\varepsilon \rightarrow 0$ at fixed ε^2/α , the discrete update may be summarized by the coarse-grained SDE

$$dx = -\frac{\eta}{\alpha} \nabla \Phi[\rho(t)](x) dt + \sqrt{2D} dW_t, \quad D \sim \frac{\varepsilon^2}{2\alpha}, \quad (8)$$

with convention-dependent numerical prefactors [15, 16]. The potential remains memory-induced and generally slowly time-dependent. In metastable regimes it can be treated as quasi-stationary on intermediate relaxation scales.

Operationally, α can be estimated from the exponential age weights in Eq. (6) or from the decay of a memory-sensitive observable such as $\Phi[\rho_n](x_n)$. The role of t is therefore not to postulate a physical clock, but to provide a dimensionless comparison parameter intrinsic to the relaxing-memory dynamics.

IV. DYNAMICAL KNOTS

The feedback between trajectory and memory can generate localized regions of repeated visitation. We call such metastable regions *dynamical knots*. They are not imposed fixed points or particles; they are persistent trajectory-memory patterns. Figure 1 shows a representative simulated trajectory in which distinct knot-like regions appear.

A knot can be identified operationally by a region $\mathcal{K} \subset \Omega$ whose memory/visitation weight

$$M_n(\mathcal{K}) = \int_{\mathcal{K}} \rho_n(x) dx \quad (9)$$

remains elevated over intervals $\Delta n \gg \alpha^{-1}$. Elevated $M_n(\mathcal{K})$ means that the memory density carries substantial weight near $\mathcal{K}$; through Eq. (4) this can generate a locally restoring drift that stabilizes the region against stochastic excursions.

In a frozen-memory approximation, knots can be represented locally by points or regions where

$$\nabla \Phi[\rho](x_\star) = 0. \quad (10)$$

Trajectory organization and knot transitions

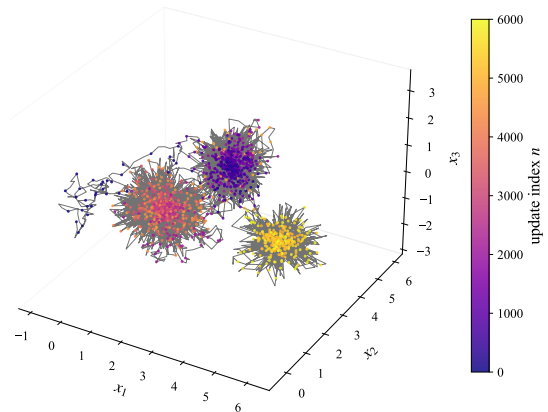


FIG. 1. Representative trajectory with multiple metastable knot-like regions. The line shows the visible trajectory, sampled points contribute to the memory density, and color denotes update index.

Linearizing the continuum drift near such a point yields an OU-type approximation

$$dx = -\gamma H(x - x_\star) dt + \sqrt{2D} dW_t, \quad \gamma = \frac{\eta}{\alpha}, \quad (11)$$

where H is the Hessian of the memory potential at $x_\star$. Local stability requires the symmetric part of H to be positive in the relevant directions [17]. If λ denotes a characteristic restoring eigenvalue, the relaxation rate scales as

$$\Gamma_{\text{rel}} \sim \frac{\eta}{\alpha} \lambda. \quad (12)$$

V. RELAXATION DIAGNOSTICS

The stability of a dynamical knot is measured by how quickly perturbations relax back into the metastable region. Rare transitions are expected in the usual metastable regime of noisy dynamics [18, 19]. Let τ_{rel} be extracted, for example, from an in-knot autocorrelation $C(t) \propto e^{-t/\tau_{\text{rel}}}$ or from the dominant relaxation eigenvalue of the linearized Fokker–Planck operator. We define

$$\Gamma_{\text{rel}} = \frac{1}{\tau_{\text{rel}}}. \quad (13)$$

Thus the diagnostic is a fitted decay scale of a coarse-grained trajectory observable, not a new microscopic parameter.

Because absolute coordinate scales on Ω are not physically fixed, it is useful to report the dimensionless combination

$$\hat{\Gamma} = \frac{\sigma^2}{D} \Gamma_{\text{rel}}. \quad (14)$$

This compares relaxation to diffusion on the coarse-graining scale and is approximately invariant under uniform coordinate rescalings $x \mapsto ax$ with $\sigma \mapsto a\sigma$. A second useful coarse-grained scale is

$$Q_{\text{rel}} = 2D\Gamma_{\text{rel}}, \quad (15)$$

which summarizes stochastic broadening and relaxation without assigning physical energy units. A later calibration may determine whether a normalized version of this quantity can be interpreted as energy-like.

In this operational sense, a mass-like proxy is associated with relaxation or confinement strength rather than with a fundamental inertial parameter. Heavier structures correspond to faster return to the metastable region and/or steeper memory-induced restoring potentials. This terminology is only an effective-description shorthand; no identification with mc^2 or a relativistic mass scale is assumed here.

VI. DISCUSSION

We have introduced a minimal stochastic dynamics in which a visible point-valued state is coupled to an explicitly relaxing memory density. The visible dynamics is generally non-Markovian, while the augmented state (x_n, ρ_n) gives a natural Markov embedding. The

same memory update compresses ordered histories and therefore supplies an intrinsic update direction. Coarse-graining on the memory-persistence scale yields the dimensionless parameter $t = \alpha n$ and, in suitable regimes, an approximate SDE/Fokker–Planck description.

Within this effective description, memory feedback can generate metastable dynamical knots. Their stability can be characterized by residence weights, relaxation rates, and dimensionless diagnostics such as $\hat{\Gamma}$. These quantities are extractable from simulation trajectories and do not require physical spacetime units.

The construction remains intentionally limited. The continuum SDE is an effective approximation rather than a proven scaling theorem, and the existence and classification of knot regimes depend on the qualitative properties of (G_σ, K) . Multiple coupled generators, emergent geometry, universal constants, quantum normal forms, and Standard-Model-like structures are natural next questions, but they are not assumptions or conclusions of the present minimal model.

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- [1] S. Weinberg. Ultraviolet divergences in quantum theories of gravitation. In S. W. Hawking and W. Israel, editors, *General Relativity: An Einstein Centenary Survey*, pages 790–831. Cambridge University Press, 1979.
 - [2] M. Reuter. Nonperturbative evolution equation for quantum gravity. *Phys. Rev. D*, 57:971–985, 1998.
 - [3] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin. Spacetime as a causal set. *Phys. Rev. Lett.*, 59:521–524, 1987.
 - [4] C. Rovelli. *Quantum Gravity*. Cambridge University Press, 2004.
 - [5] J. Polchinski. *String Theory. Volume I: An Introduction to the Bosonic String*. Cambridge University Press, 1998.
 - [6] R. Pemantle. A survey of random processes with reinforcement. *Probability Surveys*, 4:1–79, 2007.
 - [7] D. J. Amit, G. Parisi, and L. Peliti. Asymptotic behavior of the true self-avoiding walk. *Phys. Rev. B*, 27:1635–1645, 1983.
 - [8] B. Tóth and W. Werner. The true self-repelling motion. *Probab. Theory Relat. Fields*, 111:375–452, 1998.
 - [9] M. Benaïm and O. Raimond. Self-interacting diffusions. *Probability Theory and Related Fields*, 122:1–41, 2002.
 - [10] R. T. Durrett and L. C. G. Rogers. Asymptotic behavior of brownian polymers. *Probab. Theory Relat. Fields*, 92(3):337–349, 1992.
 - [11] L. Peliti and L. Pietronero. Random walks with memory. *La Rivista del Nuovo Cimento*, 10(6):1–33, 1987.
 - [12] H. Mori. Transport, collective motion, and brownian motion. *Prog. Theor. Phys.*, 33:423–455, 1965.
 - [13] R. Zwanzig. *Nonequilibrium Statistical Mechanics*. Oxford University Press, 2001.
 - [14] C. Gardiner. *Stochastic Methods*. Springer, 2009.
 - [15] B. Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. Springer, 6 edition, 2003.
 - [16] H. Risken. *The Fokker–Planck Equation: Methods of Solution and Applications*. Springer, 2 edition, 1996.
 - [17] S. Strogatz. *Nonlinear Dynamics and Chaos*. CRC Press, 2018.
 - [18] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4): 284–304, 1940.
 - [19] M. Freidlin and A. Wentzell. *Random Perturbations of Dynamical Systems*. Springer, 3 edition, 2012.